Software Quality Assurance (SQA) – 4th Week Project Task

Name: Dariya Hashmi -- DHC (2403)

Project Title: Performance & Load Testing Test Plan

Objective:

The objective of this performance test is to evaluate the **login and logout functionality** of **The Internet demo site** (https://the-internet.herokuapp.com) under multiple virtual users using JMeter. The test measures **response time, throughput, and error rate**, and verifies the system's ability to handle concurrent users.

Test Environment:

- System: Windows 10, 8GB RAM, Intel i5
- JMeter Version: 5.6.3
- Browser: Chrome 146.0
- Website Tested: The Internet Demo Site

Load Test Scenario:

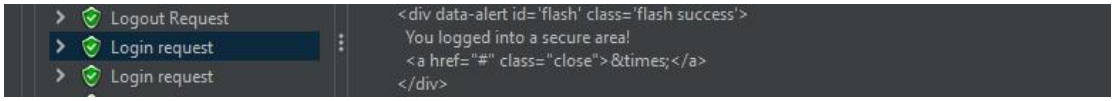
Parameter	Value
Number of Users	5
Ramp-Up Period	5 seconds
Loop Count	1
Samplers	Login → Logout
Config Element	HTTP Cookie Manager
Listeners	View Results Tree, Graph Results, Summary Report

Observations / Results :

- Login Response: "You logged into a secure area!"
- Logout Response: "You logged out of the secure area!"
- Error Rate: 0%
- Average Response Time: ~1–2 seconds
- Throughput: Stable for 5 users

Screenshots:

Fig 1: View Results Tree – Login & Logout Responses



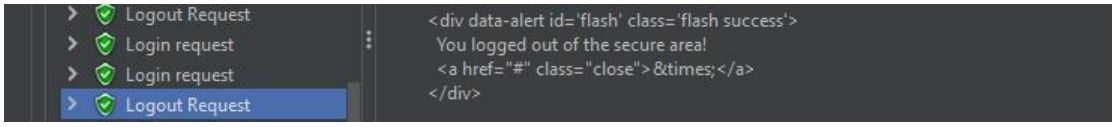


Fig 2: Summary Report – Performance Metrics

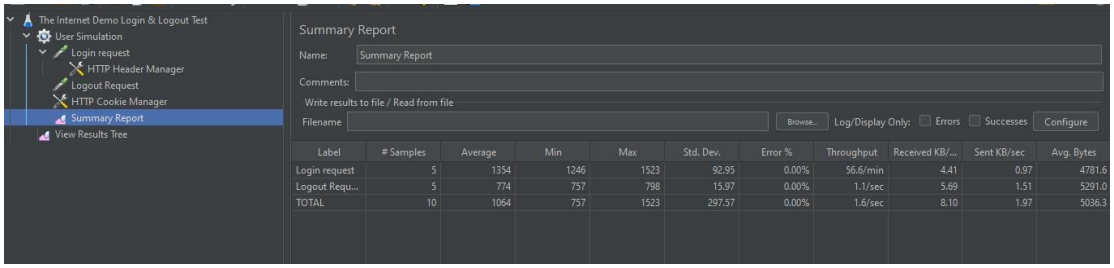
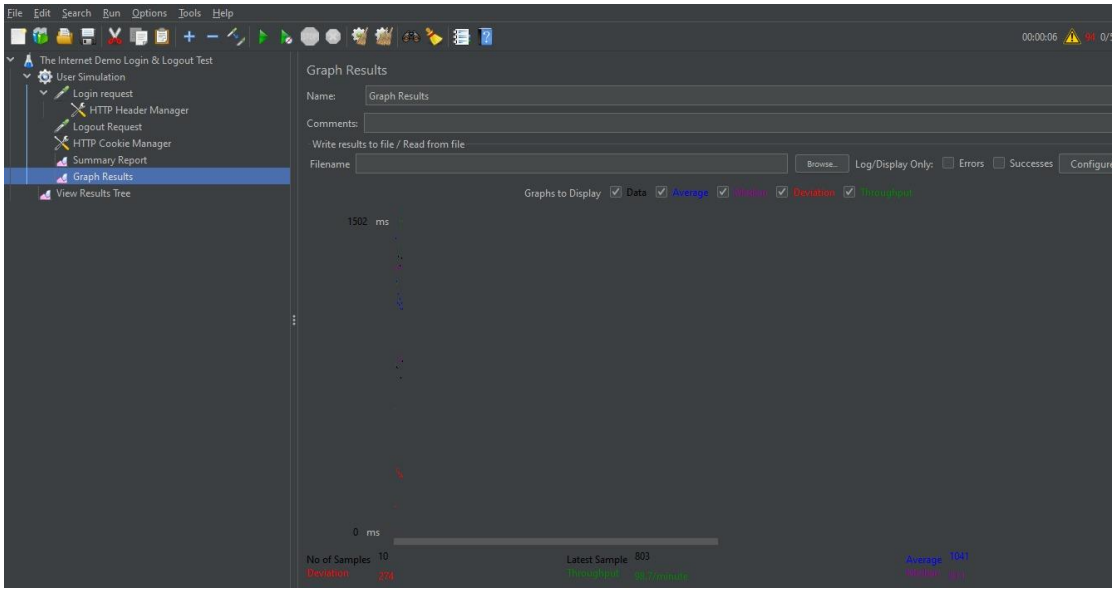


Fig 3: Graph Results – Response Time & Throughput



Conclusion & Recommendations:

- The Internet demo site successfully handled 5 concurrent users.
- Average response time remained under 2 seconds.
- No errors observed during the test.

- **Recommendation:** Gradually increase threads to 10–50 for higher load testing.

Github Repository :

https://github.com/dariyahashmi2602-source/SQA_Week4_performance-report